Q1.

Scientists no longer accept the geocentric model of the universe but it was the accepted theory for hundreds of years.

Explain why the evidence available at the time supported the geocentric model.

(3)

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(Total for question = 3 marks)

Q2.

* Figure 14 shows two ice skaters during a sequence in their performance.

Figure 14

The man stays at the same place on the ice throughout the sequence.

At the start of the sequence, the woman is moving at a constant speed around the man while the man holds her arm.

After she has gone round the man several times, the man lets go of the woman's arm.
The sequence ends a few seconds later.

Explain the motion of the woman, in terms of the forces acting and the effects on her motion, for the whole sequence.

(6)

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(Total for question = 6 marks)

Q3.

Observations of the Universe can be made using telescopes on Earth.

Explain why some telescopes are located on satellites that orbit the Earth.

(2)

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(Total for question = 2 marks)

Q4.

* The Sun is likely to remain as a main sequence star for 2000 million years.

Explain why the Sun does not explode now.

(6)

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(Total for question = 6 marks)

Q5.

The Asteroid Belt is part of our Solar System.

Vesta is an asteroid in the Asteroid Belt.

Explain why Vesta is accelerating even when it is travelling at a constant speed.

(2)

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.....

(Total for question = 2 marks)

Q6.

Which row of the table is correct for both force and velocity?

(1)

	force	velocity
☐ A	scalar	scalar
☐ B	scalar	vector
☐ C	vector	scalar
☐ D	vector	vector

(Total for question = 1 mark)

Q7.

Figure 6 shows a satellite orbiting the Earth.

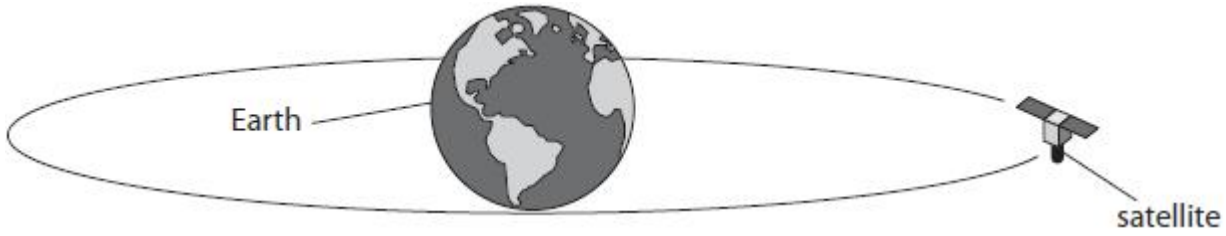

Figure 6

(i) State the name of the force that keeps the satellite in orbit around the Earth.

(1)

.....

(ii) Draw an arrow on Figure 6 to show the direction of the force acting on the satellite, that keeps the satellite in orbit around the Earth.

Label this arrow 'F'.

(1)

(Total for question = 2 marks)

Q8.

This picture shows Galileo holding a telescope.

(a) Explain the purpose of the eyepiece in a telescope.

(2)

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.....

(b) Galileo drew pictures of his observations of Jupiter.
Nowadays we can take photographs.
Suggest how photographs would have helped Galileo.

(1)

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.....

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.....

(c) The telescope collects light reflected from Jupiter.
The light has a frequency of 4.30×10^{14} Hz and a speed of 3.00×10^{8} m/s.
Calculate the wavelength of the light.

(3)

wavelength =..... m

*(d) Galileo's observations of the moons of Jupiter disproved the geocentric model.
However, these observations were not enough to prove the heliocentric model of the Solar System.
Explain why Galileo's observations disproved one model but were not enough to prove the other model.

(6)

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(Total for Question = 12 marks)

Q9.

State **two** advantages of using a camera, rather than the unaided eye, for studying stars.

(2)

1

.....

.....

2

.....

.....

Q10.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The Asteroid Belt is part of our Solar System.

Vesta is an asteroid in the Asteroid Belt.

Vesta orbits the Sun between the orbits of

(1)

☐ **A** Venus and Earth

☐ **B** Earth and Mars

☐ **C** Mars and Jupiter

☐ **D** Jupiter and Saturn

(Total for question = 1 mark)

Q11.

(i) A long time ago, scientists believed that the Earth was at the centre of the Solar System.

Evidence has since proved that the Sun is at the centre of the Solar System.
State **one other** idea about the Solar System that **has** changed over time.

(1)

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.....

(ii) Figure 4 shows data for some of the planets of the Solar System.

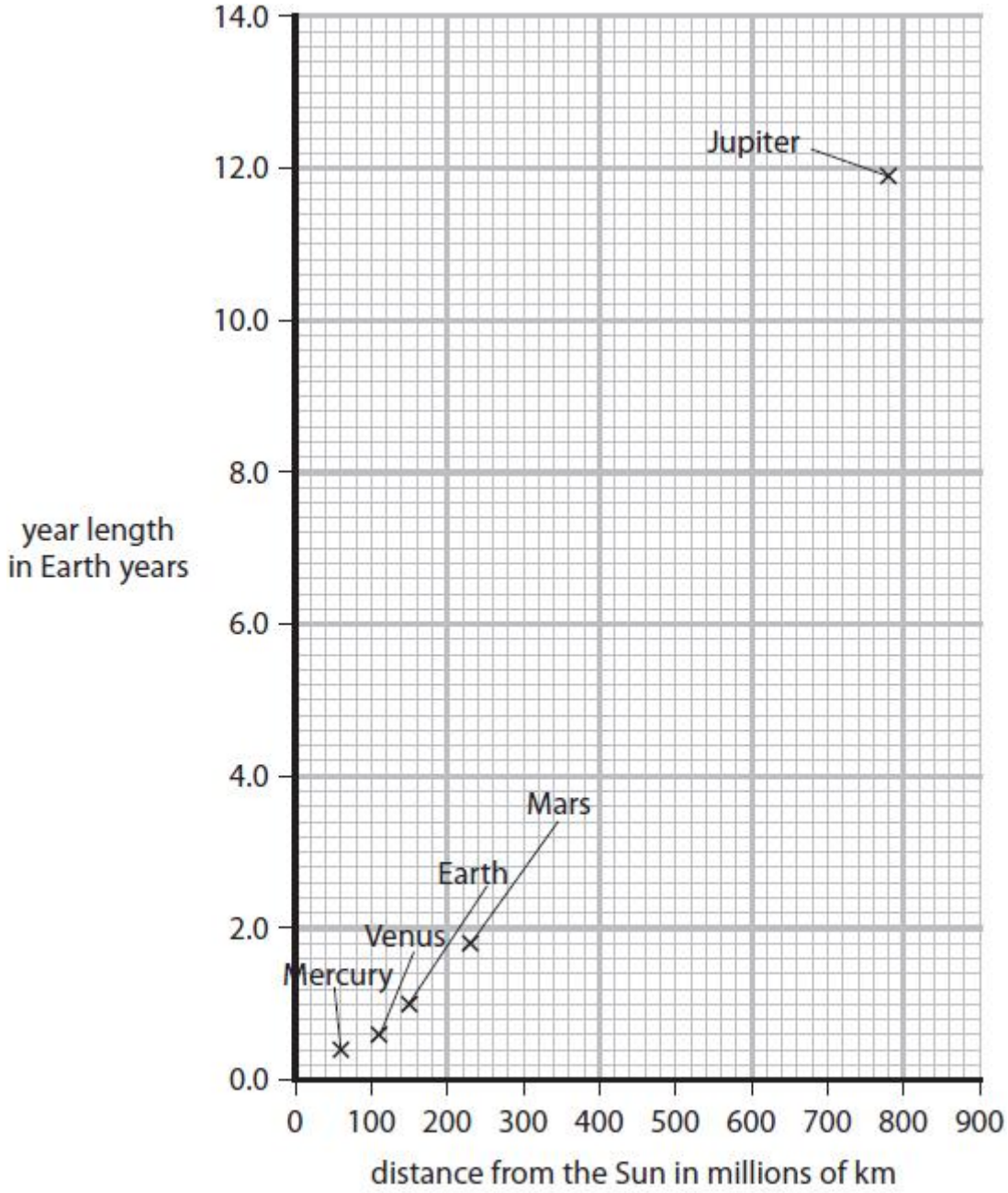

Figure 4

Ceres is an asteroid that orbits the Sun between Mars and Jupiter.
It takes Ceres 4.6 Earth years to make one orbit of the Sun.
Use the graph to estimate the distance of Ceres from the Sun.
Show your working.

(3)

distance of Ceres from the Sun = ... millions of km

(Total for question = 4 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
	An explanation that combines identification – understanding (1 mark) and reasoning/justification – understanding (2 marks): • at the time, there was only naked-eye evidence (1) • which indicated Sun/Moon/planets appear to move across the sky (1) • in the same direction, same motion each day (1)	allow valid alternatives, e.g. references to comets	(3)

Q2.

Question number	Indicative content	Mark
*	The indicative content below is not prescriptive and candidates are not required to include all of the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **A01 (6 marks)** • force needed to keep an object moving in a circle • when moving in a circle, direction of velocity changes • must be an acceleration • moving in a straight line with no resultant force at constant velocity **A02 (6 marks)** • the woman changing direction while circling the man • she is changing velocity (but not changing speed) • therefore she is accelerating • this requires a force towards the centre of her orbit • this is a centripetal force • when the man releases the woman, the centripetal force ceases • there is no resultant force on the woman (if friction from the ice can be ignored) • the woman therefore continue in a straight line • she is now travelling at a constant velocity	(6)

Level	Mark	Descriptor
	0	No awardable content
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • The explanation attempts to link and apply knowledge and understanding of scientific ideas, flawed or simplistic connections made between elements in the context of the question. (AO2)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • The explanation is mostly supported through linkage and application of knowledge and understanding of scientific ideas, some logical connections made between elements in the context of the question. (AO2)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • The explanation is supported throughout by linkage and application of knowledge and understanding of scientific ideas, logical connections made between elements in the context of the question. (AO2)

Q3.

	Answer	Additional guidance	Mark
	an explanation linking:	allow any other sensible benefit	**(2)** **AO1**
	telescopes are above Earth / in space / outside the Earth's atmosphere (1)		
	less interference from dust / clouds (1)	allow atmospheric distortion/ (light/air) pollution	
	wavelengths that are absorbed by Earth's atmosphere (microwaves / IR / UV) (1)	accept named examples of satellites or waves	
	can view the whole sky (1)		
		do not accept idea that telescope is closer to object in space	

Q4.

Question number	Indicative content	Mark
*	Answers will be credited according to the candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **A01 (6 marks)** • Gravity is an attractive force • Gravity pulls stellar matter inwards • Fusion reactions in the core release thermal energy • Thermal energy gives rise to expansion force • Sun is (currently) stable because these two forces are in equilibrium • Mass of Sun is important factor • Red supergiants explode in supernova • Sun does not have enough mass to be a red supergiant	**(6)**

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of science ideas lacks detail (A01) • Presents an explanation with some structure and coherence (A01)
Level 2	3–4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5–6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Q5.

	Answer	Additional guidance	Mark
	An explanation linking **two** from: $a = \frac{\Delta v}{t}$ (1) velocity/ acceleration is a vector (1) the direction (of Vesta/velocity) is changing (1)	 velocity is changing vector has magnitude and direction	**(2)** **AO1**

Q6.

Question Number	Answer	Mark
	D | vector | vector **is the only correct answer** ***A*** *'scalar scalar' is incorrect, both force and velocity are vectors* ***B*** *'scalar vector' is incorrect, with force being described incorrectly as a scalar* ***C*** *'vector scalar' is incorrect, with velocity being described incorrectly as a scalar*	**(1)** AO 1 1

Q7.

Question Number	Answer	Additional guidance	Mark
(i)	gravitational / centripetal (1)	accept gravity / weight/ gravitational field strength	**(1)** AO 2 1

Question Number	Answer	Additional guidance	Mark
(ii)	arrow from the satellite towards the centre of Earth - by eye (1)	direction must be clear	**(1)** AO 2 1

Q8.

Question Number	Answer	Acceptable answers	Mark
(a)	An explanation linking any two of: MP1 magnify (1) with one of: MP2 the (real) image from objective (lens) (1) MP3 to provide greater detail (1)	 enlarges / bigger ignore zooming the real image (in the telescope) / image at focal point ignore make it clearer inversion of image focuses image	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)	One sensible suggestion, such as • made recording results {easier /quicker} (1) • results more convincing to other people (1) • photograph is to scale (1)	For example, don't have to keep looking through telescope would be (better) proof / evidence (i.e. between geo- and helio-centric models) (eg multiple photographs would prove movement/orbit of moons) (photograph) is more {accurate / precise / reliable}/ can measure (relative) separations of moons (from planet) better ignore more detail/clearer/zooming	**(1)**

Question Number	Answer	Acceptable answers	Mark
(c)	Substitution (1) $3.0 \times 10^8 = 4.3 \times 10^{14} \times \lambda$ Transposition (1) $(\lambda=) \quad \frac{3.0 \times 10^8}{4.3 \times 10^{14}}$ Evaluation (1) 6.98×10^{-7} (m)	Substitution and transposition in either order Ignore triangle correct answer no working = 3 power of ten error = 2 to at least 2sf (eg 7.0/6.97.....)$\times 10^{-7}$ Ignore powers of 10 until evaluation	**(3)**

Question Number		Indicative content	Mark
QWC	*** (d)**	A description to include some of the following points description of models • geocentric • heliocentric description of one set of observations of Jupiter's moons explanation of how observation contradicts geocentric but does not prove the other NB beware that you do not reward repetitions of the question stem	**(6)**

Level	0	no rewardable material
1	1-2	• a limited explanation of the geocentric **AND** the heliocentric model such as *geocentric model said everything orbited the Earth while the other was for everything going round the Sun.* **OR** clearly explains the one model and describes Galileo's observations *eg geocentric model said everything orbited the Earth but Galileo observed that Jupiter had moons going around it* • the answer communicates ideas using simple language and uses limited scientific terminology e.g. allow confusion between geocentric and heliocentric • spelling, punctuation and grammar are used with limited accuracy
2	3-4	• a simple explanation of geocentric **AND** heliocentric models **AND** Galileo's observations of Jupiter's moons/explains heliocentric not proved e.g. *The geocentric model said everything orbited the Earth while the heliocentric was for everything orbiting the Sun. Galileo observed that Jupiter had moons orbiting around it.* • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately e.g. correct use geocentric and heliocentric • spelling, punctuation and grammar are used with some accuracy
3	5-6	• a detailed explanation of geocentric **AND** heliocentric models **AND** the role of Galileo in providing evidence against the geocentric **BUT** not enough for the heliocentric such as *The geocentric model said everything orbited the Earth while the heliocentric was for everything orbiting the Sun. Galileo's observations that Jupiter had moons orbiting arou(nd it showed that the geocentric model was wrong* ***but not*** *that Jupiter or anything else went around the Sun.* • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately e.g. distinguishes between the necessary and insufficient conditions • spelling, punctuation and grammar are used with few errors

Q9.

Question Number	Answer	Acceptable answers	Mark
	Any two suitable such as: • Measurements can be taken (1) • Permanent record/evidence (1) • Can be magnified (1) • Can detect waves outside visible part of spectrum (1) • Long exposure (to see faint objects/track objects) (1)	 Analysis/compare 'can record data' Taking photo is insufficient zoom in/show more detail can detect gamma rays, X-rays, ultraviolet, infrared Allow collect more light IGNORE better, brighter, clearer	**(2)**

Q10.

	Answer	**Mark**
	C Mars and Jupiter A, B and D are not correct because these are not the location of the asteroid belt	**(1)** **AO1**

Q11.

Question Number	Answer	Additional guidance	Mark
(i)	examples: planets have moons (1) the Earth rotates (spins) (1) planets orbit the Sun (1) Pluto is no longer a planet (1) orbits are elliptical (not circular) (1) there are more planets than previously thought (1) ours is not the only solar system (1) Earth is {round/spherical /not flat} (1) planets are not wandering stars (1)	answers must be to do with the solar system	**(1)**

Question Number	Answer	Additional guidance	Mark
(ii)	smooth curve drawn on the graph (1) horizontal line drawn from 4.6 Earth years to intercept the drawn line/curve (1) EITHER: their reading from line / curve (1) OR 430±30 (million km) (1)	accept curve up to Mars followed by a straight line plot point /draw line at year length = 4.6 reading on distance axis ± half small square from their graph award full marks for the correct answer without working	**(3)**